IT342-G5
System Integration and Architecture

System Design Document (SDD)

Project Title: SynChef

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Version: 1.0

Date: 5/27/2026
Status: Deployment

REVISION HISTORY TABLE

Version	Date	Author	Changes Made	Status
0.1	2/20/2026	Vince L. Batawang	Initial draft	Draft
0.2	2/25/2026	Vince L. Batawang	Added API specifications	Completed
0.3	3/26/2026	Vince L. Batawang	Updated database design	Completed
0.4	3/27/2026	Vince L. Batawang	Added UI/UX designs	Completed
0.5	4/20/2026	Vince L. Batawang	Incorporated feedback	Completed
0.6	5/04/2026	Vince L. Batawang	Final review and corrections	Final
1	5/20/2026	Vince L. Batawang	Baseline version for development	Approved

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EXECUTIVE SUMMARY

1.1 Project Overview

SynChef is a cross-platform smart culinary guidance system that transforms static recipes into an interactive, execution-focused cooking experience.

The project combines a ***Java Spring Boot backend (Maven***), a ***Vite/React-based*** web frontend, an Android mobile client, and containerized deployments (**Docker**) to provide synchronized step orchestration, adaptive scaling, and culturally aware recipe recommendations.

SynChef prioritizes in-kitchen assistance with real-time timers, parallel task coordination, and minimal cognitive load for home cooks.

1.2 Objectives

1. Enable real-time orchestration of parallel cooking tasks and synchronized timers to simplify multi-step recipes.
2. Implement dynamic ingredient scaling and time adjustments to match household sizes and reduce food waste.
3. Provide culturally contextual recipe discovery and rule-based substitution suggestions to increase accessibility.
4. Deliver a step-focused “Focus Mode” UI to guide users through individual steps with clear, actionable prompts.
5. Expose RESTful APIs for recipes, steps, timers, scaling logic, and user data to support web and mobile clients.
6. Establish a modular, container-friendly architecture to enable future AI features (substitutions, timing optimization, personalization).

1.3 Scope

Included Features:

- User accounts and authentication: registration, login, and profile management.
- Recipe browsing and discovery: filtering by cuisine, category, and dietary constraints.
- Parallel timer orchestration: coordinated timers and step synchronization across devices.
- Dynamic ingredient & time scaling: adjust quantities and durations based on servings and cookware.
- Step-by-step Focus Mode: simplified, sequential instructions with progress state.
- Rule-based cultural substitutions: MVP substitution suggestions for local ingredients.
- REST API services: endpoints for recipes, steps, timers, scaling, and user preferences.
- Relational data storage: production-ready database (Postgres or equivalent).
- Responsive web frontend (Vite/React) and Android client with Dockerized deployment artifacts.

Excluded Features:

- Full AI-driven personalization and recommendations (deferred to future phases).
- Voice assistant or hands-free voice control integration.
- Grocery delivery or external commerce integrations.
- Computer-vision-based cooking detection or real-time camera analysis.
- Social media sharing, community feeds, or in-app monetization/payments.
- Deep sensor integrations (IoT stove/smart cookware) in the initial release.

1.0 INTRODUCTION

1.1 Purpose

This document serves as the comprehensive design specification for the SynChef system. It provides detailed functional requirements, architectural decisions, API communication structure, database design considerations, and an implementation roadmap to guide the development process and ensure all system components integrate seamlessly.

By defining these specifications early, the document minimizes integration risks, prevents design inconsistencies, and establishes a structured foundation for delivering a reliable, scalable, and user-centered guided cooking platform.

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

2.1 Project Overview

Project Name: SynChef
Domain: Smart Food Technology / Guided Cooking System
Primary Users:

- Home cooks
- Learners
- Food Enthusiasts

Problem Statement: Users lack an interactive system that can guide real-time cooking, coordinate simultaneous tasks, and adapt recipes dynamically to reduce waste and complexity.
Solution: SynChef converts recipes into synchronized, execution-driven workflows that

guide users in real time with timers, scaled ingredients, and culturally contextual discovery.

2.2 Core User Journey

Journey 1: Guided Cooking Experience

1. User logs into the system.
2. Selects a recipe from the Flavor Map.
3. Inputs number of servings (Pax).
4. System recalculates ingredients and timing.
5. User enters Focus Mode.
6. Parallel timers guide task execution.
7. User completes cooking with synchronized alerts.

Journey 2: Exploring Global Cuisine

1. User opens Flavor Map
2. Selects continent → country
3. Views culturally authentic recipes
4. Reads ingredient background and preparation method
5. Saves recipe to personal collection

Journey 3: Adjusting Meal Portions

1. User selects a recipe
2. Changes serving size
3. System dynamically updates:
    - Ingredient quantities
    - Cooking durations
    - Required steps
4. User proceeds with updated workflow

2.3 Feature List (MoSCoW)

MUST HAVE

1. User authentication (register/login/logout)
2. Recipe management and retrieval
3. Parallel timer execution engine
4. Dynamic ingredient scaling
5. Step-by-step Focus Mode
6. Country-based recipe categorization
7. RESTful API communication

SHOULD HAVE

1. Ingredient substitution suggestions
2. Recipe filtering by dietary category

3. Saved recipes functionality
4. Responsive UI animations for transitions

COULD HAVE

1. AI-based timing optimization
2. Personalized recommendations
3. Multilingual instructions

WON'T HAVE

1. Full AI automation
2. Grocery ordering integration
3. Smart kitchen device integration
4. Voice-controlled cooking

2.4 Detailed Feature Specifications

Feature: User Authentication

- **Screens**: Registration, Login
- **Fields**: Name, Email, Password, Confirm Password
- **Validation**: Email format validation, password length (min. 8 characters), uniqueness of account
- **Security**: JWT-based authentication, password hashing using Django authentication system
- **API Endpoints**:

  POST /api/auth/register
  POST /api/auth/login
  POST /api/auth/logout

Feature: Recipe Catalog Management

- **Screens**: Recipe Listing, Recipe Detail
- **Display**: Recipe image, name, country of origin, preparation time, servings
- **Search**: By recipe name, category, or country
- **Filtering**: Vegan, Dessert, Main Dish, etc.
- **API Endpoints:**

GET /api/recipes
GET /api/recipes/{id}
GET /api/recipes/search
GET /api/recipes/country/{country}

Feature: Global Flavor Map Navigation

- **Screens**: Interactive Map View, Country Selection View
- **Display**: Continent → Country navigation with cuisine highlights
- **Functions**: Load culturally relevant recipes based on selected region
- **Purpose**: Promote culinary education and cultural exploration
- **API Endpoints:**

GET /api/countries
GET /api/recipes/country/{country}

Feature: Adaptive Parallel Timer System

- **Screens**: Cooking Mode, Active Timer Dashboard
- **Functions**: Run multiple cooking timers simultaneously, synchronize task completion
- **Display**: Countdown timers per step, alerts for next action
- **Logic**: Backend schedules step overlap to optimize cooking workflow
- **API Endpoints:**

GET /api/recipes/timer-sequence/{id}
POST /api/timers/start
POST /api/timers/update

Feature: Dynamic Ingredient Scaling

- **Screens**: Recipe Detail, Serving Adjustment Panel
- **Fields**: Number of servings (Pax input)
- **Functions**: Automatically recompute ingredient quantities and cooking duration
- **Precision Handling**: Decimal-based calculations for accurate scaling
- **API Endpoints:**

POST /api/recipes/scale
GET /api/recipes/{id}

Feature: Progressive Step-by-Step Focus Mode

- **Screens**: Focus Cooking Interface

- **Display**: One instruction visible at a time with navigation controls
- **Functions**: Prevent user distraction, guide sequential execution
- **Integration**: Syncs with timer engine for automated progression
- **API Endpoints:**

    GET /api/recipes/{id}/steps

Feature: Ingredient Substitution Assistance (Rule-Based MVP)

- **Screens**: Ingredient Information Panel
- **Functions**: Suggest alternative ingredients based on availability or region
- **Logic**: Predefined substitution mappings stored in database
- **Purpose**: Improve accessibility and reduce food waste
- **API Endpoints:**

    GET /api/ingredients/substitutions/{ingredient}

Feature: Saved Recipes / Personal Collection

- **Screens**: User Dashboard, Saved Recipes List
- **Functions**: Bookmark recipes for future cooking
- **Persistence**: Stored per authenticated user
- **API Endpoints:**

    GET /api/users/saved
    POST /api/users/saved/{recipeId}
    DELETE /api/users/saved/{recipeId}

2.5 Acceptance Criteria

AC-1: Successful Recipe Execution

- Given a user selects a recipe
- When Focus Mode is activated
- Then the system must guide the user step-by-step
- And synchronize timers automatically
- And notify when each action must begin.

AC-2: Ingredient Scaling Accuracy

- Given a recipe designed for 4 servings

- When the user changes to 2 servings
- Then all ingredient values must adjust proportionally
- And updated measurements must display instantly.

AC-3: Parallel Task Coordination

- Given a recipe contains overlapping steps
- When cooking begins
- Then timers must run simultaneously
- And alerts must ensure synchronized completion.

AC-4: Cultural Recipe Discovery

- Given a user selects a country
- When recipes load
- Then the system must display cuisine-specific dishes
- And associated preparation context.

3.0 NON-FUNCTIONAL REQUIREMENTS

1.1 Performance Requirements

- API response time: ≤ 2 seconds for 95% of requests
- Web page load time: ≤ 3 seconds on broadband
- Mobile app cold start: ≤ 3 seconds
- Support 100 concurrent users
- Database queries complete within 500ms

1.2 Security Requirements

- HTTPS for all communications
- JWT token authentication
- Password hashing with bcrypt (salt rounds = 12)
- SQL injection prevention
- XSS protection
- Rate limiting: 100 requests/minute per IP
- Admin endpoints require role verification

1.3 Compatibility Requirements

- **Web Browsers:** Chrome, Firefox, Safari, Edge (latest 2 versions)
- **Android:** API Level 24+ (Android 7.0+)
- **Screen Sizes:** Mobile (360px+), Tablet (768px+), Desktop (1024px+)
- **Operating Systems:** Windows 10+, macOS 10.15+, Linux Ubuntu 20.04+

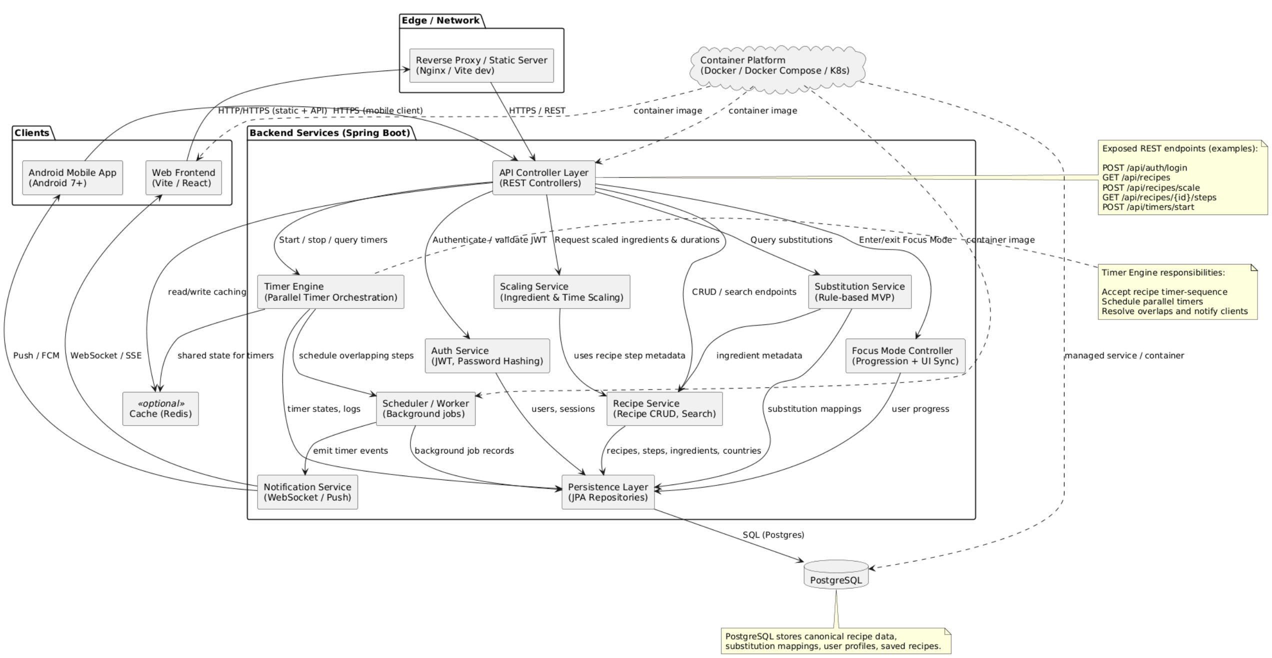
1.4 Usability Requirements

- Complete first purchase within 5 minutes for new users
- WCAG 2.1 Level AA compliance for web
- Consistent navigation across all pages
- Clear error messages with recovery options
- Touch targets minimum 44x44px on mobile
- Keyboard navigation support

4.0 SYSTEM ARCHITECTURE

4.1 Component Diagram

Note: This should be a component diagram



Technology Stack:

- **Backend:** Java 17, Spring Boot 3.x, Spring Security, Spring Data JPA
- **Database:** PostgreSQL 14+
- **Web Frontend:** React 18, TypeScript, Tailwind CSS, Axios
- **Mobile:** Kotlin, Jetpack Compose, Retrofit, Room
- **Build Tools:** Maven (Backend), npm/yarn (Web), Gradle (Android)
- **Deployment:** Railway/Heroku (Backend), Vercel/Netlify (Web), APK (Mobile)

5.0 API CONTRACT & COMMUNICATION

5.1 API Standards

Base URL	https://[server_hostname]:[port]/api/v1
Format	JSON for all requests/responses
Authentication	Bearer token (JWT) in Authorization header

Response Structure	{
	"success": boolean,
	"data": object|null,
	"error": {
	"code": string,
	"message": string,
	"details": object|null
	},
	"timestamp": string
	}

5.2 Endpoint Specifications

Authentication Endpoints

User Registration

Description	User Registration
API URL	/auth/register
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None

Request Payload	{
	“email”: <email>,
	“password”: <password>,
	“firstname”: <firstname>,
	“lastname”: <lastname>,
	}

Response Structure	{
	"success": boolean,
	"data": {
	user {
	“email”: <email>,
	“firstname”: <firstname>,
	“lastname”: <lastname>,
	},
	“accessToken”: <token>,
	“refreshToken”: <token>,
	},
	"error": {
	"code": string,
	"message": string,
	"details": object|null
	},
	"timestamp": string
	}

User Login

Description	User Login
API URL	/auth/login
HTTP Method	POST
Format	JSON for all requests/responses
Authentication	None

Request Payload	{
	“email”: <email>,
	“password”: <password>,
	}

Response Structure	{
	"success": boolean,
	"data": {
	“user”: {
	“email”: <email>,
	“firstname”: <firstname>,
	“lastname”: <lastname>,
	“role”: <role>,
	},
	“accessToken”: <token>,
	“refreshToken”: <token>,
	},
	"error": {
	"code": string,
	"message": string,
	"details": object|null
	},
	"timestamp": string
	}

5.3 Error Handling

HTTP Status Codes

- 200 OK - Successful request
- 201 Created - Resource created
- 400 Bad Request - Invalid input
- 401 Unauthorized - Authentication required/failed
- 403 Forbidden - Insufficient permissions
- 404 Not Found - Resource doesn't exist
- 409 Conflict - Duplicate resource
- 500 Internal Server Error - Server error

Error Code Examples

Example 1	json { "success": false,

	"data": null,
	"error": {
	"code": "AUTH-001",
	"message": "Invalid credentials",
	"details": "Email or password is incorrect"
	},
	"timestamp": "2024-01-28T10:30:00Z"
	}

Example 2	{
	"success": false,
	"data": null,
	"error": {
	"code": "VALID-001",
	"message": "Validation failed",
	"details": {
	"email": "Email is required",
	"password": "Must be at least 8 characters"
	}
	},
	"timestamp": "2024-01-28T10:30:00Z"
	}

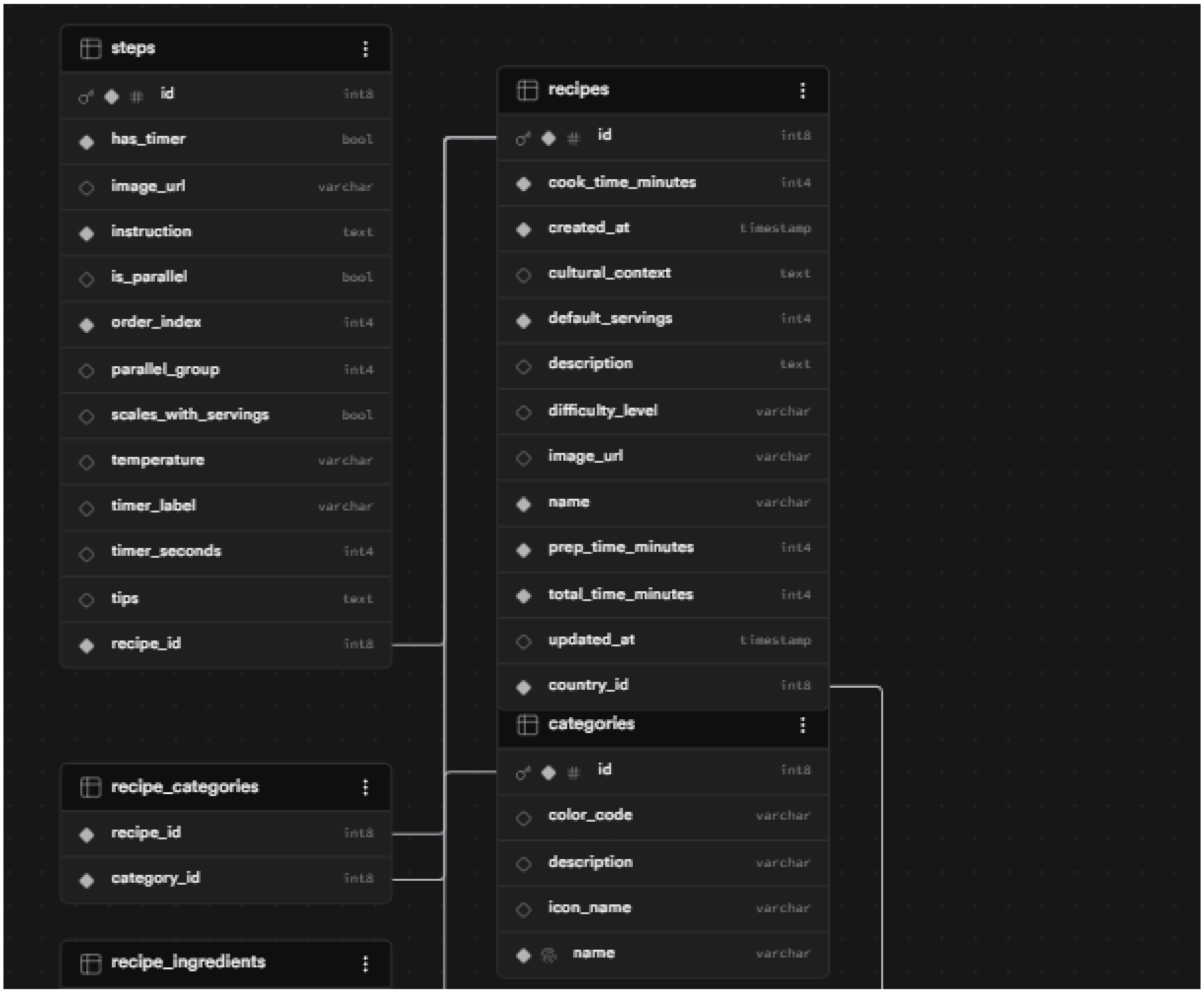
Common Error Codes

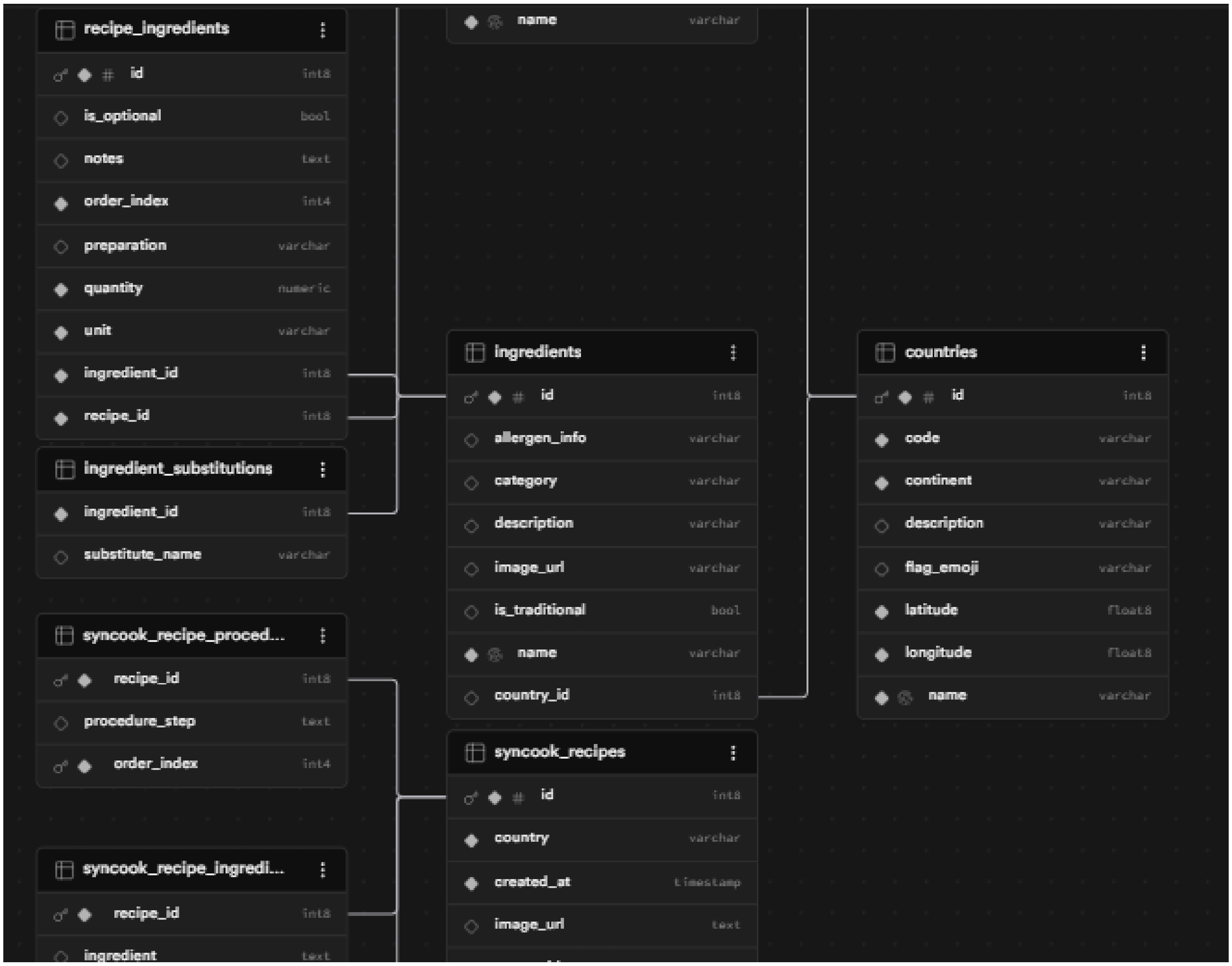
- AUTH-001: Invalid credentials
- AUTH-002: Token expired
- AUTH-003: Insufficient permissions
- VALID-001: Validation failed
- DB-001: Resource not found
- DB-002: Duplicate entry
- BUSINESS-001: Insufficient stock
- SYSTEM-001: Internal server error

6.0 DATABASE DESIGN

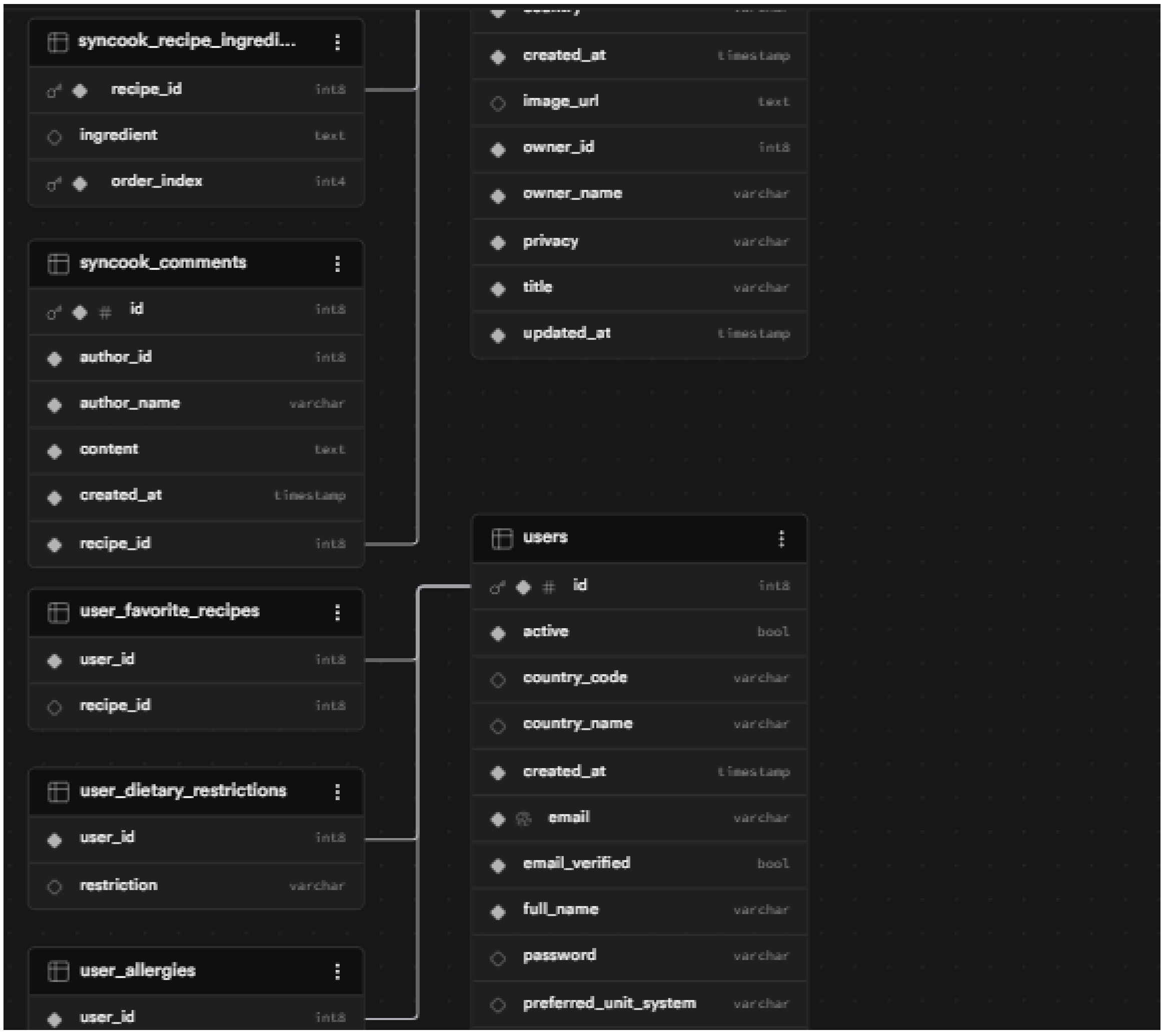
6.1 Entity Relationship Diagram

Note: This should be an ERD (Database Schema)

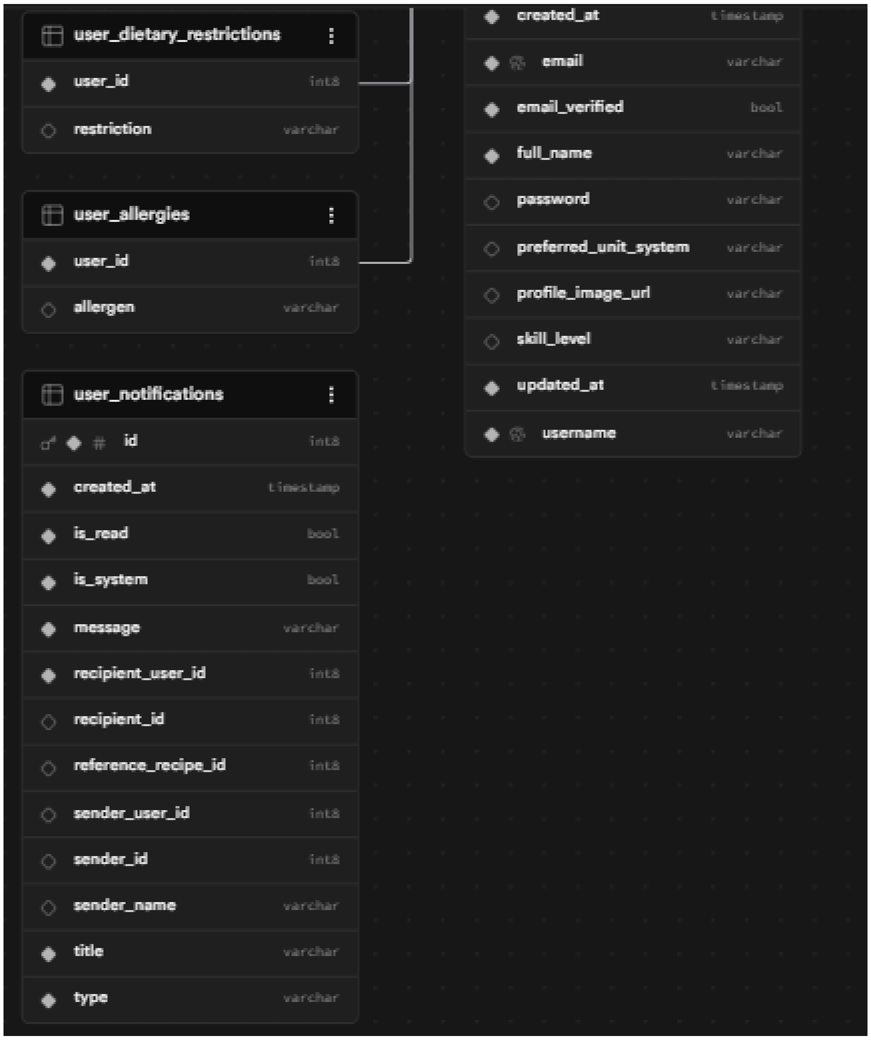

steps
id int8
has_timer bool
image_url varchar
instruction text
is_parallel bool
order_index int4
parallel_group int4
scales_with_servings bool
temperature varchar
timer_label varchar
timer_seconds int4
tips text
recipe_id int8
recipes
id int8
cook_time_minutes int4
created_at timestamp
cultural_context text
default_servings int4
description text
difficulty_level varchar
image_url varchar
name varchar
prep_time_minutes int4
total_time_minutes int4
updated_at timestamp
country_id int8
categories
id int8
color_code varchar
description varchar
icon_name varchar
name varchar
recipe_categories
recipe_id int8
category_id int8
recipe_ingredients



name varchar
recipe_ingredients
id int8
is_optional bool
notes text
order_index int4
preparation varchar
quantity numeric
unit varchar
ingredient_id int8
recipe_id int8
ingredient_substitutions
ingredient_id int8
substitute_name varchar
syncook_recipe_proced...
recipe_id int8
procedure_step text
order_index int4
syncook_recipe_ingredi...
recipe_id int8
ingredient text
ingredients
id int8
allergen_info varchar
category varchar
description varchar
image_url varchar
is_traditional bool
name varchar
country_id int8
syncook_recipes
id int8
country varchar
created_at timestamp
image_url text
countries
id int8
code varchar
continent varchar
description varchar
flag_emoji varchar
latitude float8
longitude float8
name varchar



created_at
timestamp
image_url
text
owner_id
int8
owner_name
varchar
privacy
varchar
title
varchar
updated_at
timestamp
syncook_recipe_ingredi...
recipe_id
int8
ingredient
text
order_index
int4
syncook_comments
id
int8
author_id
int8
author_name
varchar
content
text
created_at
timestamp
recipe_id
int8
user_favorite_recipes
user_id
int8
recipe_id
int8
user_dietary_restrictions
user_id
int8
restriction
varchar
user_allergies
user_id
int8
users
id
int8
active
bool
country_code
varchar
country_name
varchar
created_at
timestamp
email
varchar
email_verified
bool
full_name
varchar
password
varchar
preferred_unit_system
varchar



Detailed Relationships:

- **One-to-One:** User ↔ Cart (Each user has exactly one cart)
- **One-to-Many:** User → Orders (User can have multiple orders)
- **One-to-Many:** Cart → CartItems (Cart contains multiple items)
- **One-to-Many:** Order → OrderItems (Order contains multiple items)
- **Many-to-One:** CartItems → Product (Items reference products)
- **Many-to-One:** OrderItems → Product (Items reference products)

Key Tables:

1. **users** - User accounts and authentication
2. **products** - Product catalog information
3. **carts** - Shopping cart per user
4. **cart_items** - Items in shopping cart
5. **orders** - Customer orders
6. **order_items** - Items in each order
7. **refresh_tokens** - JWT refresh tokens

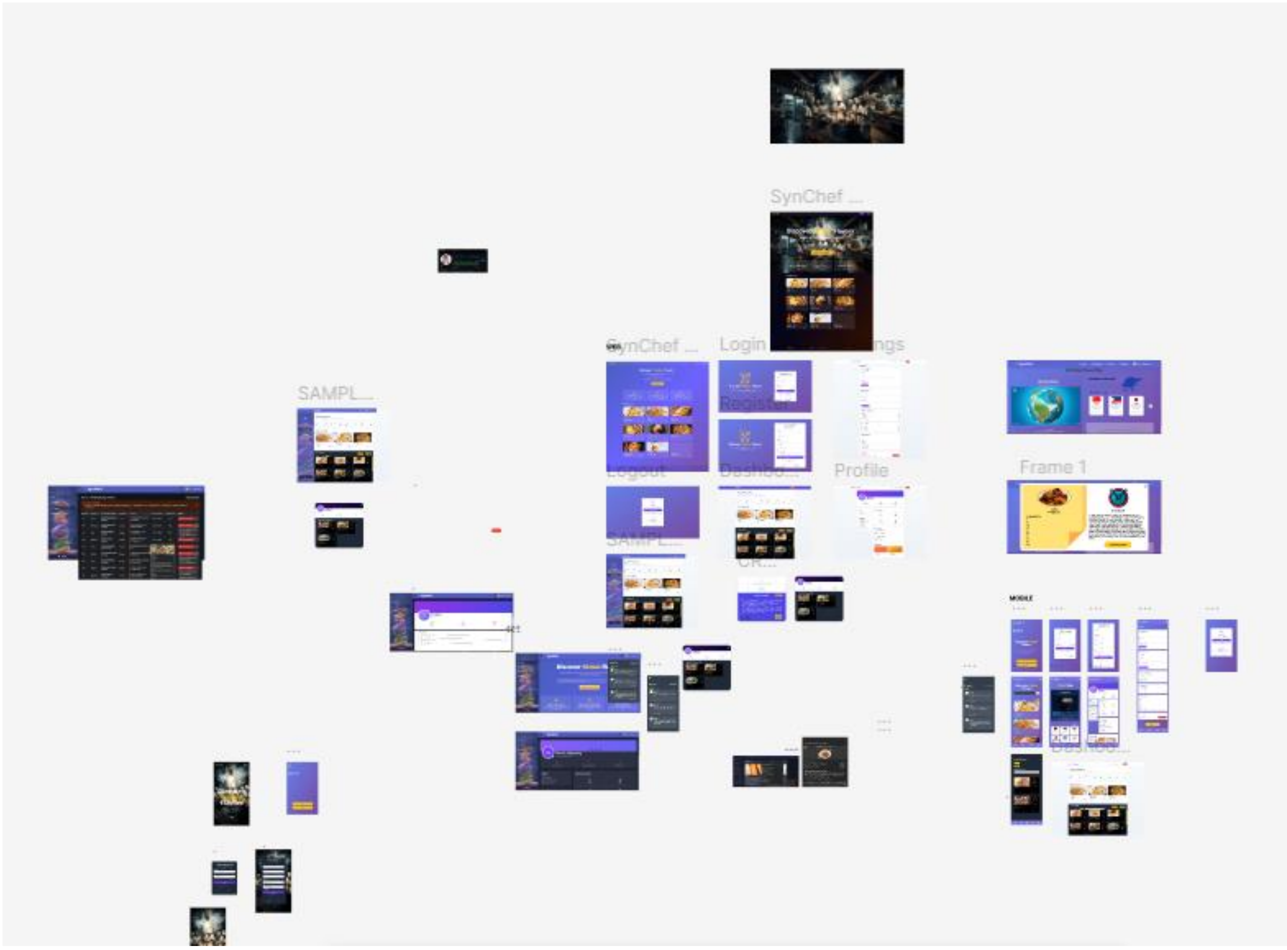
Table Structure Summary:

- **users:** id, email, password_hash, full_name, role, created_at
- **products:** id, name, description, price, stock, image_url, category
- **carts:** id, user_id, created_at
- **cart_items:** id, cart_id, product_id, quantity
- **orders:** id, order_number, user_id, total, status, shipping_address
- **order_items:** id, order_id, product_id, product_name, quantity, price

7.0 UI/UX DESIGN

7.1 Web Application Wireframes

Note: This should be wireframes from Figma



Homepage (Product Listing)

Header: [Logo] [Search Bar] [Cart Icon] [User Menu]

Content: Product Grid (3 columns desktop)

Each Product Card: Image, Name, Price, "Add to Cart" button

Footer: Links, Copyright

Product Detail Page

Back Button

Product Image (large)

Product Name and Price

Description

Quantity Selector (1-10)

"Add to Cart" and "Buy Now" buttons

Product Specifications

Shopping Cart Page

Cart Title

List of Cart Items (Image, Name, Quantity, Price, Remove)

Order Summary: Subtotal, Shipping, Tax, Total

"Continue Shopping" and "Proceed to Checkout" buttons

Checkout Page

Shipping Address Form

Order Review (Items, Prices, Totals)

"Place Order" button

Terms and Conditions note

Admin Dashboard

Sidebar Navigation: Dashboard, Products, Orders, Users

Product Management: Add New button, Product list with Edit/Delete

Order Management: Order list with status filters

7.2 Mobile Application Wireframes

Note: This should be wireframes from Figma

Bottom Navigation

[🏠 Home] [🔍 Search] [🛒 Cart] [👤 Profile]

Home Screen

Search Bar

Product Grid (2 columns)

Swipe gestures for quick actions

Pull to refresh

Product Detail Screen

Back arrow

Product image (swipeable gallery)

Product info

Quantity selector

"Add to Cart" fixed bottom button

Cart Screen

Edit mode for quantity updates

Swipe to remove items

Order summary sticky bottom

Checkout button

Checkout Flow

Step indicator: Cart → Shipping → Payment → Confirm

Address form (auto-complete)

Order summary

Place order button

Mobile-Specific Features:

- Touch-optimized buttons (min 44x44px)
- Gesture support (swipe, pull-to-refresh)
- Offline caching for product images
- Bottom navigation for main actions
- Simplified forms for mobile input

Design System:

- **Colors:** Primary (#2563EB), Secondary (#7C3AED), Success (#10B981), Error (#EF4444)
- **Typography:** Inter font family, responsive sizing
- **Spacing:** 8px grid system
- **Components:** Consistent buttons, inputs, cards, modals
- **Responsive:** Mobile-first approach, breakpoints at 640px, 768px, 1024px

8.0 PLAN

8.1 Project Timeline

Phase 1: Planning & Design (Week 1-2)

Week 1: Requirements & Architecture

Day 1-2: Project setup and documentation

Day 3-4: Complete FRS and NFR

Day 5-7: System architecture design

Week 2: Detailed Design

Day 1-2: API specification

Day 3-4: Database design

Day 5-6: UI/UX wireframes

Day 7: Implementation plan finalization

Phase 2: Backend Development (Week 3-4)

Week 3: Foundation

Day 1: Spring Boot setup with dependencies

Day 2: Database configuration and entities

Day 3: JWT authentication implementation

Day 4: User management endpoints

Day 5: Product CRUD operations

Week 4: Core Features

Day 1: Cart functionality

Day 2: Order management

Day 3: Search and filtering

Day 4: Error handling and validation

Day 5: API documentation and testing

Phase 3: Web Application (Week 5-6)

Week 5: Frontend Foundation

Day 1: React setup with TypeScript

Day 2: Authentication pages (login, register)

Day 3: Product listing page

Day 4: Product detail page

Day 5: Shopping cart implementation

Week 6: Complete Web Features

Day 1: Checkout flow

Day 2: Order history and confirmation

Day 3: Admin dashboard

Day 4: Responsive design polish

Day 5: API integration and testing

Phase 4: Mobile Application (Week 7-8)

Week 7: Android Foundation

Day 1: Android Studio setup and project structure

Day 2: Authentication screens

Day 3: Product browsing

Day 4: Shopping cart

Day 5: API service layer

Week 8: Complete Mobile App

Day 1: Checkout flow

Day 2: Order management

Day 3: UI polish and animations

Day 4: Testing on emulator/device

Day 5: APK generation and documentation

Phase 5: Integration & Deployment (Week 9-10)

Week 9: Integration Testing

Day 1: End-to-end testing across platforms

Day 2: Bug fixes and optimization

Day 3: Security review

Day 4: Performance testing

Day 5: Documentation updates

Week 10: Deployment

Day 1: Backend deployment (Railway/Heroku)

Day 2: Web app deployment (Vercel/Netlify)

Day 3: Mobile APK distribution

Day 4: Final testing

Day 5: Project submission

Milestones:

- **M1 (End Week 2):** All design documents complete
- **M2 (End Week 4):** Backend API fully functional
- **M3 (End Week 6):** Web application complete
- **M4 (End Week 8):** Mobile application complete
- **M5 (End Week 10):** Full system deployed and integrated

Critical Path:

1. Authentication system (Week 3)
2. Product catalog API (Week 3-4)
3. Shopping cart functionality (Week 4)
4. Checkout process (Week 6)
5. Cross-platform testing (Week 9)

Risk Mitigation:

- Start with simplest working version of each feature
- Test integration points early and often
- Keep backup of working versions
- Focus on core functionality before enhancements